Experiment (5)
MOSFET IV Characteristics

Objectives

- To study the IV curve of a MOSFET.
- To experimentally evaluate the threshold voltage of a MOSFET.
- To graphically find the operating point of a MOSFET circuit.

Pre-lab Work

Analyze the DC circuit below. Assume the transistor is in saturation and find expressions for the drain current and the drain-source voltage. Ignore channel length modulation.

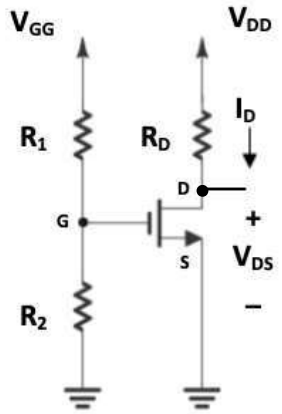


Figure 1: A MOSFET DC Circuit.

Solution

Equipment

Table 1. Equipment for experiment 5.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	Resistors 1 kΩ (1), 1 MΩ (2)			
3	NMOS transistor (1)	2N7000/7002	http://pdf.datasheetcatalog.com/datasheet/calogic/2N4351.PDF	
4	Multi-meter (1)			
5	Probes and cables			

Experimental Procedure

Part (1): MOSFET IV Characteristics.

1. Connect the circuit in figure 1 with R_D=1 kΩ and R_1= R_2=1 MΩ. Pay a special attention to how you connect the terminals of the MOSFET. Refer to the data sheet to identify the

transistor terminals. Make sure the Body terminal (if exists) is connected to the source terminal.
2. Set V_{GG} to a constant voltage of +8 V while connecting V_{DD} to a variable DC voltage source.
3. The IV characteristics (I_D versus V_{DS}) is obtained by slowly varying V_{DD} (from 0 to +10 V) and measuring the voltage V_{DS} and V_{R_D} at each setting. Use table 2 to enter your results.
4. Repeat step 3 for another value of V_{GG}, say +10 V. Record your results in table 2.

Part (2): Graphical Analysis of a DC MOSFET Circuit.

1. Reconsider the circuit in figure 1.
2. The load line equation for this circuit is $I_D = \frac{V_{DD}-V_{DS}}{R_D}$. It is the line connecting the two points $(0,\frac{V_{DD}}{R_D})$ and $(V_{DD},0)$ on the IV curve.

Part (3): MOSFET Threshold Voltage.

Produce a procedure to find the threshold voltage of the transistor using the circuit in figure 1. Write the steps below and implement them.

1.

2.

3.

4.

Results and Analysis

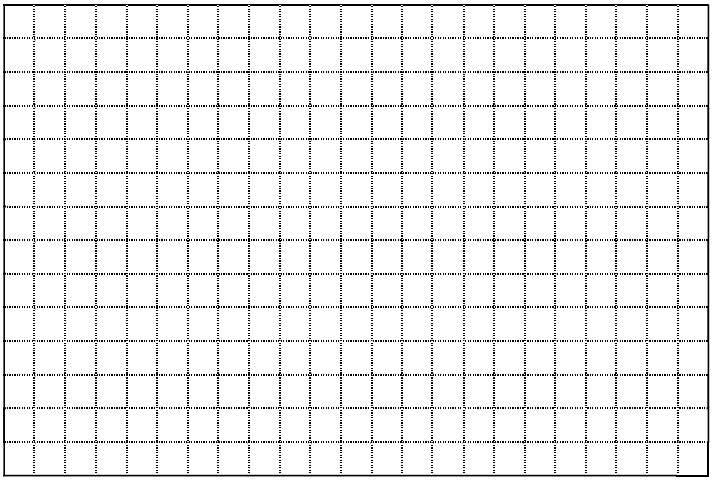
Part (1): MOSFET IV Characteristics.

1. Calculate V_{GS} and the current I_D values in table 2, for both values of V_{GG}.
2. Use graph 1 to sketch the IV curves for the two V_{GS} values.
3. Comment on the IV curves that you obtained. Can you identify the transistor modes of operation?

Answer:

Table 2. IV Characteristics Measurements.

	V_{GG} = 8V, V_{GS} =			V_{GG} = 10V, V_{GS} =		
V_{DD} (V)	V_{DS} (V)	V_{R_D} (V)	I_D (mA)	V_{DS} (V)	V_{R_D} (V)	I_D (mA)
0.0						
1.0						
2.0						
3.0						
4.0						
5.0						
6.0						
7.0						
8.0						
9.0						
10.0						



Graph 1: MOSFET IV Curves.

4. Is the drain current constant in the saturation region of the IV curve? Explain why or why not.

Answer:

Part (2): Graphical Analysis of a DC MOS Circuit.

1. Draw the load line of the circuit in figure 1 with V_{DD} and V_{GG} both connected to +10 V. Draw the line on top of the IV curves in graph 1.
2. Read the drain current value and the drain-source voltage (V_{DS}) value from the intersection point in graph 1. Record the results in table 3.
3. Use the results in table 2 for the experimental V_{DS} and I_D.

Table 3. Graphical Solution Results.

V_{DD} = 10 V and V_{GG} = 10 V					
V_{DS} (V) Experimental value	V_{DS} (V) Graphical value	% Difference	I_D (mA) Experimental value	I_D (mA) Graphical value	% Difference

4. Comment on your results explaining the sources of discrepancy.

Answers:

Part (3): MOSFET Threshold Voltage.

1. Comment on your result regarding the threshold voltage as compared to the data sheet value.
2. Comment on the procedure that you used to find the threshold voltage. How can it be improved?
3. Can you estimate the value of the constant k_n for this transistor? Show your work.

Answers:

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name		Name	
ID		ID		ID		ID	
Section							

N-Channel 60-V (D-S) MOSFET

PRODUCT SUMMARY				
Part Number	**$V_{(BR)DSS}$ Min (V)**	**$r_{DS(on)}$ Max (Ω)**	**$V_{GS(th)}$ (V)**	**I_D (A)**
2N7000	60	5 @ V_{GS} = 10 V	0.8 to 3	0.2
2N7002		7.5 @ V_{GS} = 10 V	1 to 2.5	0.115
VQ1000J		5.5 @ V_{GS} = 10 V	0.8 to 2.5	0.225
VQ1000P		5.5 @ V_{GS} = 10 V	0.8 to 2.5	0.225
BS170		5 @ V_{GS} = 10 V	0.8 to 3	0.5

FEATURES

- Low On-Resistance: 2.5 Ω
- Low Threshold: 2.1 V
- Low Input Capacitance: 22 pF
- Fast Switching Speed: 7 ns
- Low Input and Output Leakage

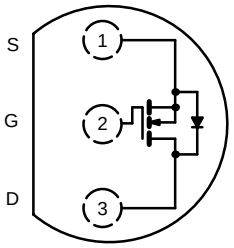
BENEFITS

- Low Offset Voltage
- Low-Voltage Operation
- Easily Driven Without Buffer
- High-Speed Circuits
- Low Error Voltage

APPLICATIONS

- Direct Logic-Level Interface: TTL/CMOS
- Drivers: Relays, Solenoids, Lamps, Hammers, Displays, Memories, Transistors, etc.
- Battery Operated Systems
- Solid-State Relays

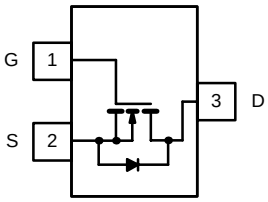
TO-226AA
(TO-92)



Top View

2N7000

TO-236
(SOT-23)

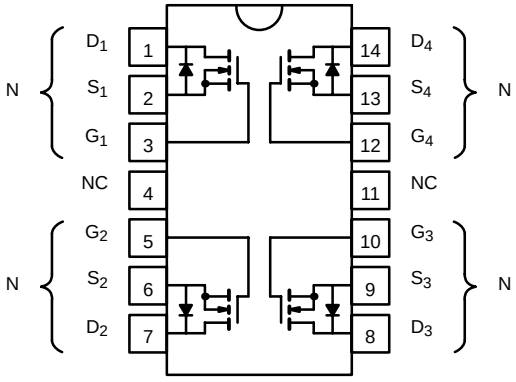


Top View

Marking Code: 72*wll*

72 = Part Number Code for 2N7002
w = Week Code
ll = Lot Traceability

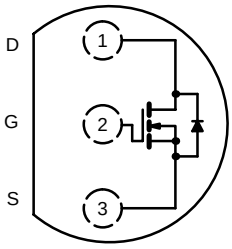
Dual-In-Line



Top View

Plastic: VQ1000J
Sidebraze: VQ1000P

TO-92-18RM
(TO-18 Lead Form)



Top View

BS170

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C UNLESS OTHERWISE NOTED)

Parameter		Symbol	2N7000	2N7002	Single VQ1000J	Single VQ1000P	Total Quad VQ1000J/P	BS170	Unit
Drain-Source Voltage		V_{DS}	60	60	60	60		60	V
Gate-Source Voltage—Non-Repetitive		V_{GSM}	±40	±40	±30			±25	V
Gate-Source Voltage—Continuous		V_{GS}	±20	±20	±20	±20		±20	V
Continuous Drain Current (T_J = 150°C)	T_A= 25°C	I_D	0.2	0.115	0.225	0.225		0.5	A
	T_A= 100°C		0.13	0.073	0.14	0.14		0.175	A
Pulsed Drain Current[a]		I_{DM}	0.5	0.8	1	1			A
Power Dissipation	T_A= 25°C	P_D	0.4	0.2	1.3	1.3	2	0.83	W
	T_A= 100°C		0.16	0.08	0.52	0.52	0.8		W
Thermal Resistance, Junction-to-Ambient		R_{thJA}	312.5	625	96	96	62.5	156	°C/W
Operating Junction and Storage Temperature Range		T_J, T_{stg}	–55 to 150						°C

Notes
a. Pulse width limited by maximum junction temperature.
b. $t_p \leq 50$ μs.

SPECIFICATIONS—2N7000 AND 2N7002 (T_A = 25°C UNLESS OTHERWISE NOTED)

Parameter	Symbol	Test Conditions	Typ[a]	Limits 2N7000 Min	Limits 2N7000 Max	Limits 2N7002 Min	Limits 2N7002 Max	Unit
Static								
Drain-Source Breakdown Voltage	$V_{(BR)DSS}$	V_{GS} = 0 V, I_D = 10 μA	70	60		60		V
Gate-Threshold Voltage	$V_{GS(th)}$	V_{DS} = V_{GS}, I_D = 1 mA	2.1	0.8	3			V
		V_{DS} = V_{GS}, I_D = 0.25 mA	2.0			1	2.5	V
Gate-Body Leakage	I_{GSS}	V_{DS} = 0 V, V_{GS} = ±15 V			±10			nA
		V_{DS} = 0 V, V_{GS} = ±20 V					±100	nA
Zero Gate Voltage Drain Current	I_{DSS}	V_{DS} = 48 V, V_{GS} = 0 V			1			μA
		T_C = 125°C			1000			μA
		V_{DS} = 60 V, V_{GS} = 0 V					1	μA
		T_C = 125°C					500	μA
On-State Drain Current[b]	$I_{D(on)}$	V_{DS} = 10 V, V_{GS} = 4.5 V	0.35	0.075				A
		V_{DS} = 7.5 V, V_{GS} = 10 V	1			0.5		A
Drain-Source On-Resistance[b]	$r_{DS(on)}$	V_{GS} = 4.5 V, I_D = 0.075 A	4.5		5.3			Ω
		V_{GS} = 5 V, I_D = 0.05 A	3.2				7.5	Ω
		T_C = 125°C	5.8				13.5	Ω
		V_{GS} = 10 V, I_D = 0.5 A	2.4		5		7.5	Ω
		T_J = 125°C	4.4		9		13.5	Ω
Forward Transconductance[b]	g_{fs}	V_{DS} = 10 V, I_D = 0.2 A		100		80		mS
Common Source Output Conductance[b]	g_{os}	V_{DS} = 5 V, I_D = 0.05 A	0.5					mS
Dynamic								
Input Capacitance	C_{iss}	V_{DS} = 25 V, V_{GS} = 0 V f = 1 MHz	22		60		50	pF
Output Capacitance	C_{oss}		11		25		25	pF
Reverse Transfer Capacitance	C_{rss}		2		5		5	pF

SPECIFICATIONS—2N7000 AND 2N7002 (T_A = 25°C UNLESS OTHERWISE NOTED)

Parameter	Symbol	Test Conditions	Typ[a]	Limits 2N7000 Min	Limits 2N7000 Max	Limits 2N7002 Min	Limits 2N7002 Max	Unit
Switching[d]								
Turn-On Time	t_{ON}	V_{DD} = 15 V, R_L = 25 Ω I_D ≅ 0.5 A, V_{GEN} = 10 V, R_G = 25 Ω	7		10			ns
Turn-Off Time	t_{OFF}		7		10			
Turn-On Time	t_{ON}	V_{DD} = 30 V, R_L = 150 Ω I_D ≅ 0.2 A, V_{GEN} = 10 V, R_G = 25 Ω	7				20	
Turn-Off Time	t_{OFF}		11				20	

SPECIFICATIONS—VQ1000J/P AND BS170 (T_A = 25°C UNLESS OTHERWISE NOTED)

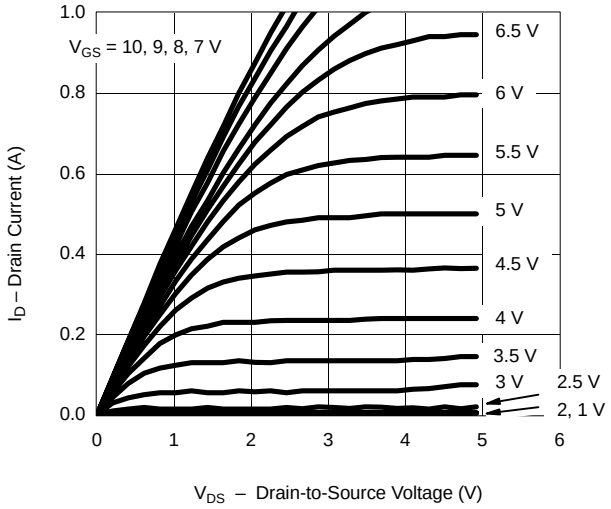
Parameter	Symbol	Test Conditions	Typ[a]	Limits VQ1000J/P Min	Limits VQ1000J/P Max	Limits BS170 Min	Limits BS170 Max	Unit
Static								
Drain-Source Breakdown Voltage	$V_{(BR)DSS}$	V_{GS} = 0 V, I_D = 100 μA	70	60		60		V
Gate-Threshold Voltage	$V_{GS(th)}$	V_{DS} = V_{GS}, I_D = 1 mA	2.1	0.8	2.5	0.8	3	
Gate-Body Leakage	I_{GSS}	V_{DS} = 0 V, V_{GS} = ±10 V			±100			nA
		T_J = 125°C			±500			
		V_{DS} = 0 V, V_{GS} = ±15 V					±10	
Zero Gate Voltage Drain Current	I_{DSS}	V_{DS} = 25 V, V_{GS} = 0 V					0.5	μA
		V_{DS} = 48 V, V_{GS} = 0 V, T_J = 125°C			500			
		V_{DS} = 60 V, V_{GS} = 0 V			10			
On-State Drain Current[b]	$I_{D(on)}$	V_{DS} = 10 V, V_{GS} = 10 V	1	0.5				A
Drain-Source On-Resistance[b]	$r_{DS(on)}$	V_{GS} = 5 V, I_D = 0.2 A	4		7.5			Ω
		V_{GS} = 10 V, I_D = 0.2 A	2.3				5	
		V_{GS} = 10 V, I_D = 0.3 A	2.3		5.5			
		T_J = 125°C	4.2		7.6			
Forward Transconductance[b]	g_{fs}	V_{DS} = 10 V, I_D = 0.2 A				100		mS
		V_{DS} = 10 V, I_D = 0.5 A		100				
Common Source Output Conductance[b]	g_{os}	V_{DS} =5 V, I_D = 0.05 A	0.5					
Dynamic								
Input Capacitance	C_{iss}	V_{DS} =25 V, V_{GS} = 0 V f = 1 MHz	22		60		60	pF
Output Capacitance	C_{oss}		11		25			
Reverse Transfer Capacitance	C_{rss}		2		5			
Switching[d]								
Turn-On Time	t_{ON}	V_{DD} = 15 V, R_L = 23 Ω I_D ≅ 0.6 A, V_{GEN} = 10 V, R_G = 25 Ω	7		10			ns
Turn-Off Time	t_{OFF}		7		10			
Turn-On Time	t_{ON}	V_{DD} = 25 V, R_L = 125 Ω I_D ≅ 0.2 A, V_{GEN} = 10 V, R_G = 25 Ω	7				10	
Turn-Off Time	t_{OFF}		7				10	

Notes
a. For DESIGN AID ONLY, not subject to production testing.
b. Pulse test: PW ≤ 80 μs duty cycle ≤ 1%.
c. This parameter not registered with JEDEC.
d. Switching time is essentially independent of operating temperature.

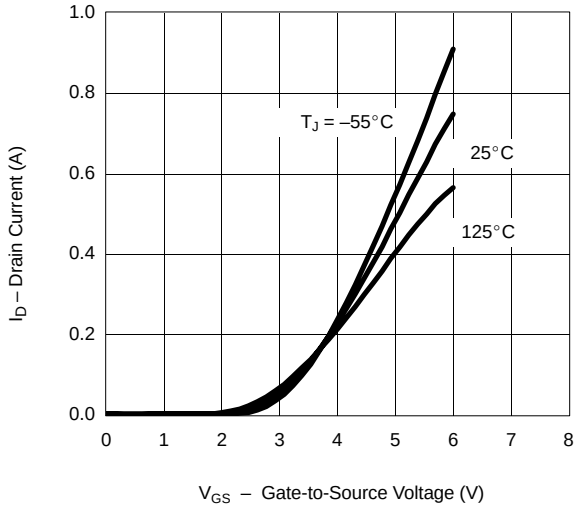
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TYPICAL CHARACTERISTICS (T_A = 25°C UNLESS OTHERWISE NOTED)

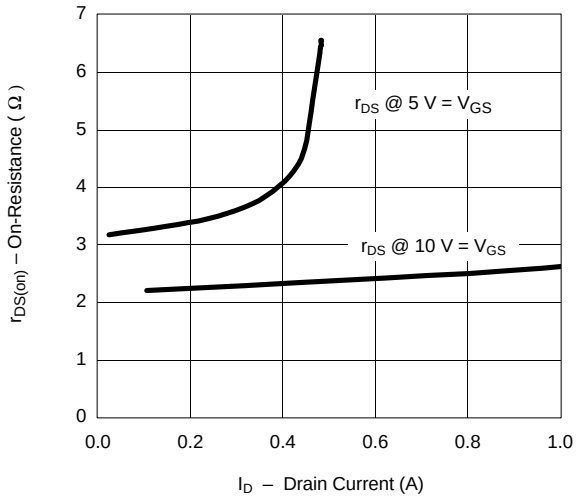
Output Characteristics



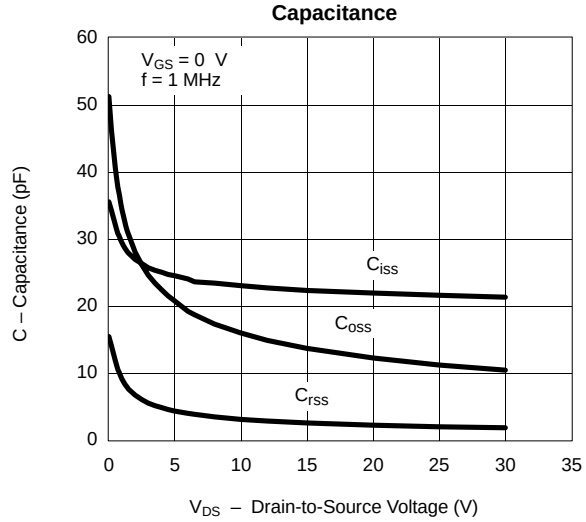
Transfer Characteristics



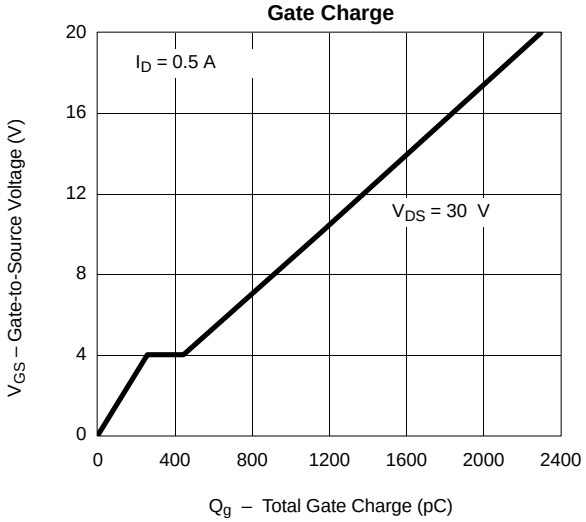
On-Resistance vs. Drain Current



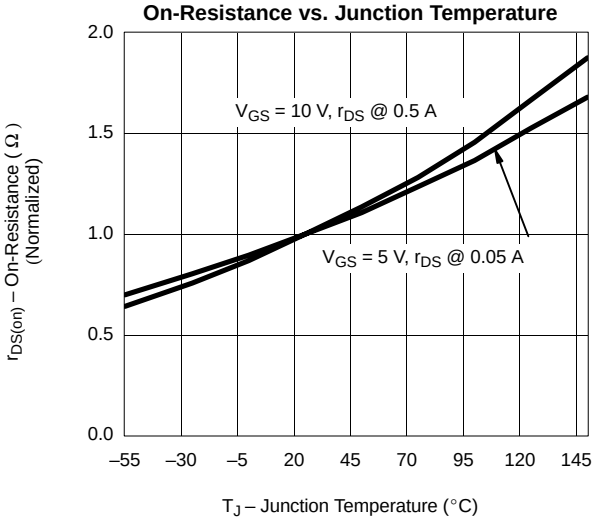
Capacitance



Gate Charge

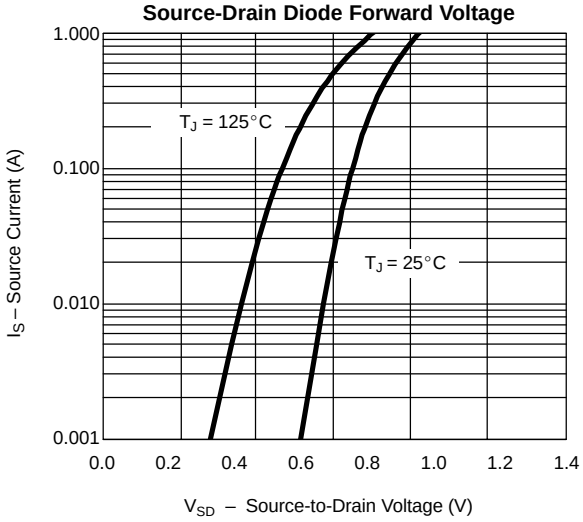


On-Resistance vs. Junction Temperature

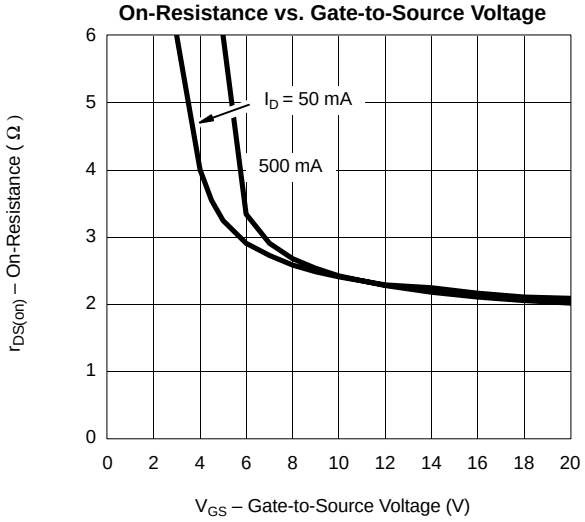


TYPICAL CHARACTERISTICS (T_A = 25°C UNLESS OTHERWISE NOTED)

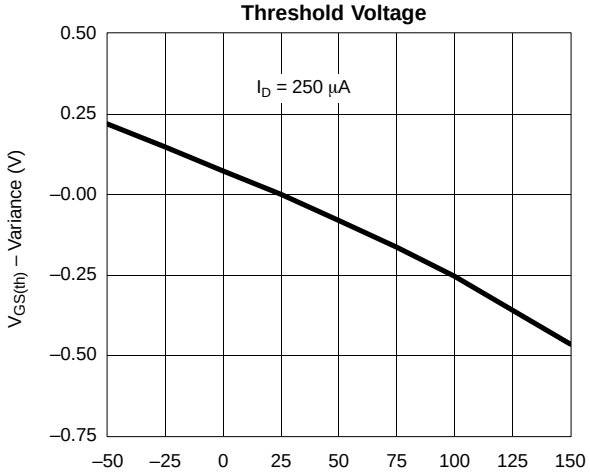
Source-Drain Diode Forward Voltage



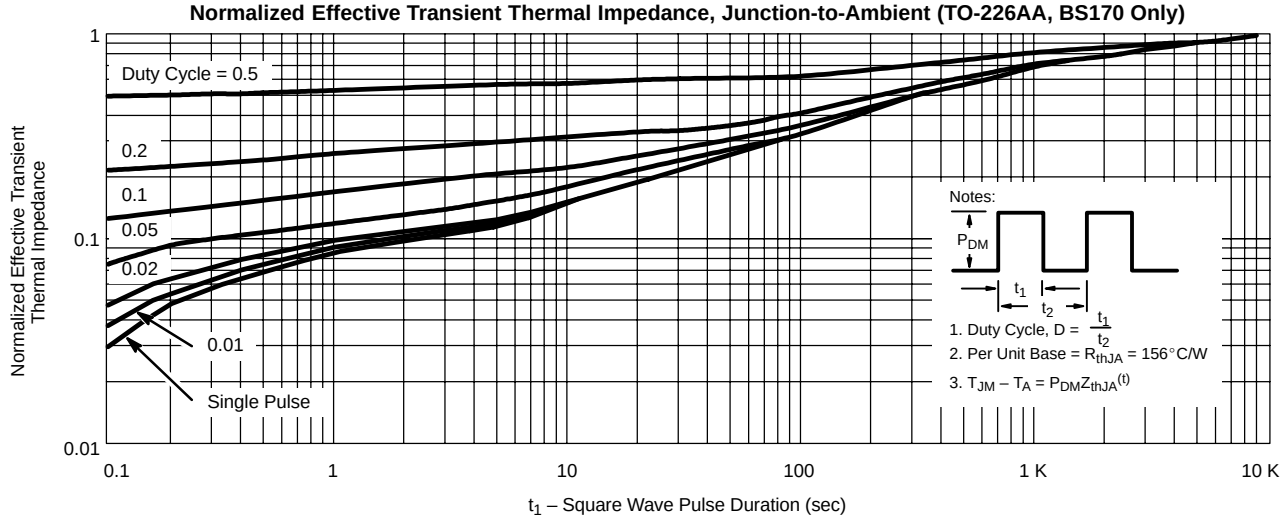
On-Resistance vs. Gate-to-Source Voltage



Threshold Voltage



Normalized Effective Transient Thermal Impedance, Junction-to-Ambient (TO-226AA, BS170 Only)



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